

BRIGHTLEARN AI & DATA SKILLS

Building a Retail Sales Data Agent on Databricks

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Table of Contents

1 Introduction & Objective	4
2 Dataset Description	5
3 Step-by-Step Evidence (Steps 1–5)	6
Step 1: Upload the Dataset	6
Step 2: Review the Dataset	7
Step 3: Prepare the Table	8
Step 4: Create the Data Agent	9
Step 5: Write the Agent Instructions	10
4 Agent Instructions	11
Section 1: Role And Purpose	11
Section 2: The Data You Work With	11
Section 3: Question Types You Support	11
Section 4: How To Structure Every Response	11
Section 5: Trend Analysis Rules	12
Section 6: Data Quality Rules	12
Section 7: Handling Ambiguous Questions	12
Section 8: Recommendation Rules	12
Section 9: Communication Style	12
5 The 10 Test Questions & Answers	13
6 Validation of 3 Answers	21
Validation 1	21
Validation 2	21
Validation 3	22
7 Key Insights	23
Insight 1: Sales Performance	23
Insight 2: Top-Performing Category & Products	24
Insight 3: Customer Behaviour	24
Insight 3.1 Gender Spending Patterns	24
Insight 3.2 Age Group Contribution	26
Insight 3.3 Spend Tier Distribution	27
Insight 4 Crosscut Analysis	29
8 Business Recommendations	30
9 Conclusion	32
What Worked Well	32
What Was Challenging	32

What I Would Change 32

Final Reflection 32

1 Introduction & Objective

This project documents the end-to-end process of building a Retail Sales Data Agent on Databricks from uploading the raw dataset to asking real business questions and validating the answers.

The goal was to create an AI-powered agent that lets a store manager, retail business owner, or executive ask questions about shop performance in plain English and receive clear, data-backed answers in seconds. No SQL required. No waiting for a report.

Why this matters:

- Retail businesses generate large volumes of transaction data that often goes underused.
- Store managers rarely have time or technical skills to run queries themselves.
- A well-instructed Data Agent bridges that gap it becomes an always-on analyst that knows your data, your business rules, and your audience.

What this document covers:

- How the retail_sales dataset was uploaded, reviewed, and prepared in Databricks.
- How the Data Agent was created and connected to the dataset.
- The agent instructions written for this specific retail context.
- Ten original business questions tested against the agent.
- Three answers independently validated against the source data.
- Key insights uncovered and business recommendations derived from the analysis.
- A reflection on what worked, what was challenging, and what could be improved.

2 Dataset Description

The dataset used in this project is a CSV file of retail sales transactions. It was loaded into Databricks as a table named `retail_sales`. Here is what each column contains:

Column Name	Data Type	What It Means
Transaction ID	String	A unique identifier for every sales transaction. Used to count transactions and avoid double-counting.
Date	Date	The date the sale took place. Used for daily, monthly, quarterly, and seasonal trend analysis.
Customer ID	String	A unique identifier for each customer. Used to track repeat purchases and customer-level behaviour.
Gender	String	The gender of the customer (Male / Female). Used for gender-based purchasing pattern analysis.
Age	Integer	The age of the customer at the time of purchase. Used to build age-group segments and analyse spending by life stage.
Product Category	String	The category of the product purchased (e.g. Electronics, Clothing, Beauty). Used for product performance analysis.
Quantity	Integer	The number of units bought in a single transaction. Used to calculate volume of goods sold.
Price per Unit	Decimal	The selling price of a single unit in ZAR. Used to calculate total revenue and average unit price.
Total Amount	Decimal	The total value of the transaction (Quantity x Price per Unit). This is the primary revenue metric used throughout the analysis.

What this data allows us to analyse:

- Sales performance: total revenue, units sold, average order value, trends over time.
- Customer behaviour: spending patterns by age group, gender, and spend tier.
- Product performance: which categories generate the most revenue and volume.
- Time-based trends: daily, weekly, monthly, quarterly, and seasonal patterns.
- Crosscuts: combinations such as which category sells best to which age group or gender.

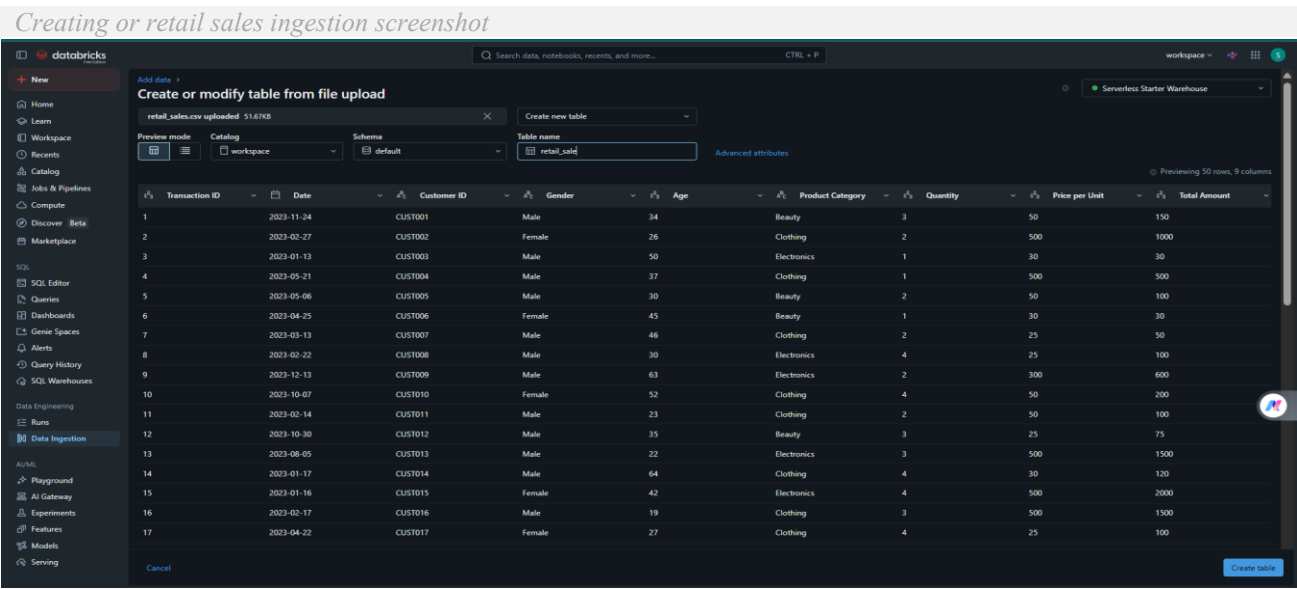
3 Step-by-Step Evidence (Steps 1–5)

Every step in this project was captured with screenshots. The evidence below describes what was done at each stage and what the screenshot shows. Screenshots are labelled by step number for easy reference in your submission.

Step 1: Upload the Dataset

The retail sales CSV file was uploaded into Databricks using the Data tab. The file was loaded using Create Table > Upload File. The row count and file size were confirmed to match the original CSV before proceeding.

Screenshot evidence: Screenshot shows: the uploaded dataset in the Databricks file browser, the preview of the first rows, and the confirmed file location.

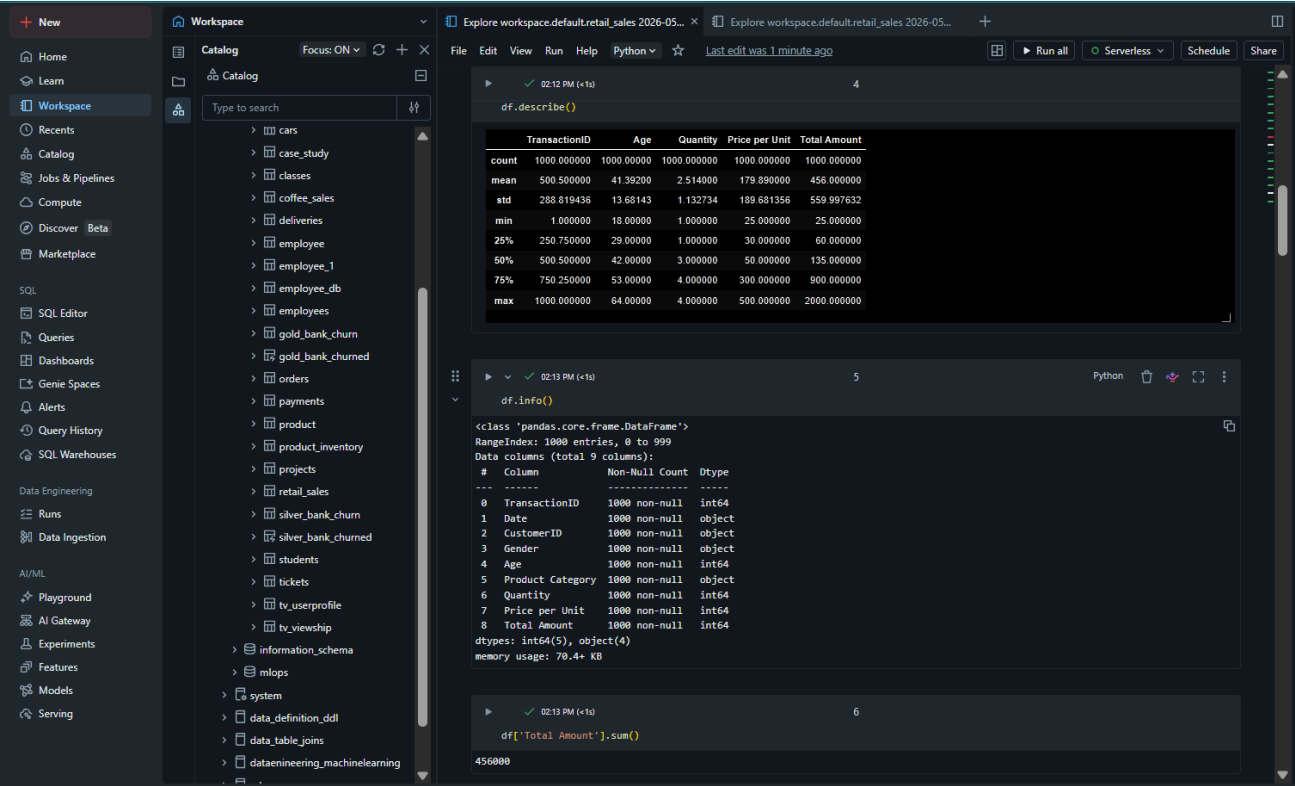


Step 2: Review the Dataset

Each column was inspected for data type, sample values, and potential quality issues. `df.info()` was run to confirm column types and `df.describe()` was used to get summary statistics. Notes were made on the age range, any missing values, and the spread of product categories.

Screenshot evidence: Screenshot shows: the table `info()`(nulls, data types, columns names and total number of columns), the `describe()` summary statistics, and a preview of the column list with data types visible.

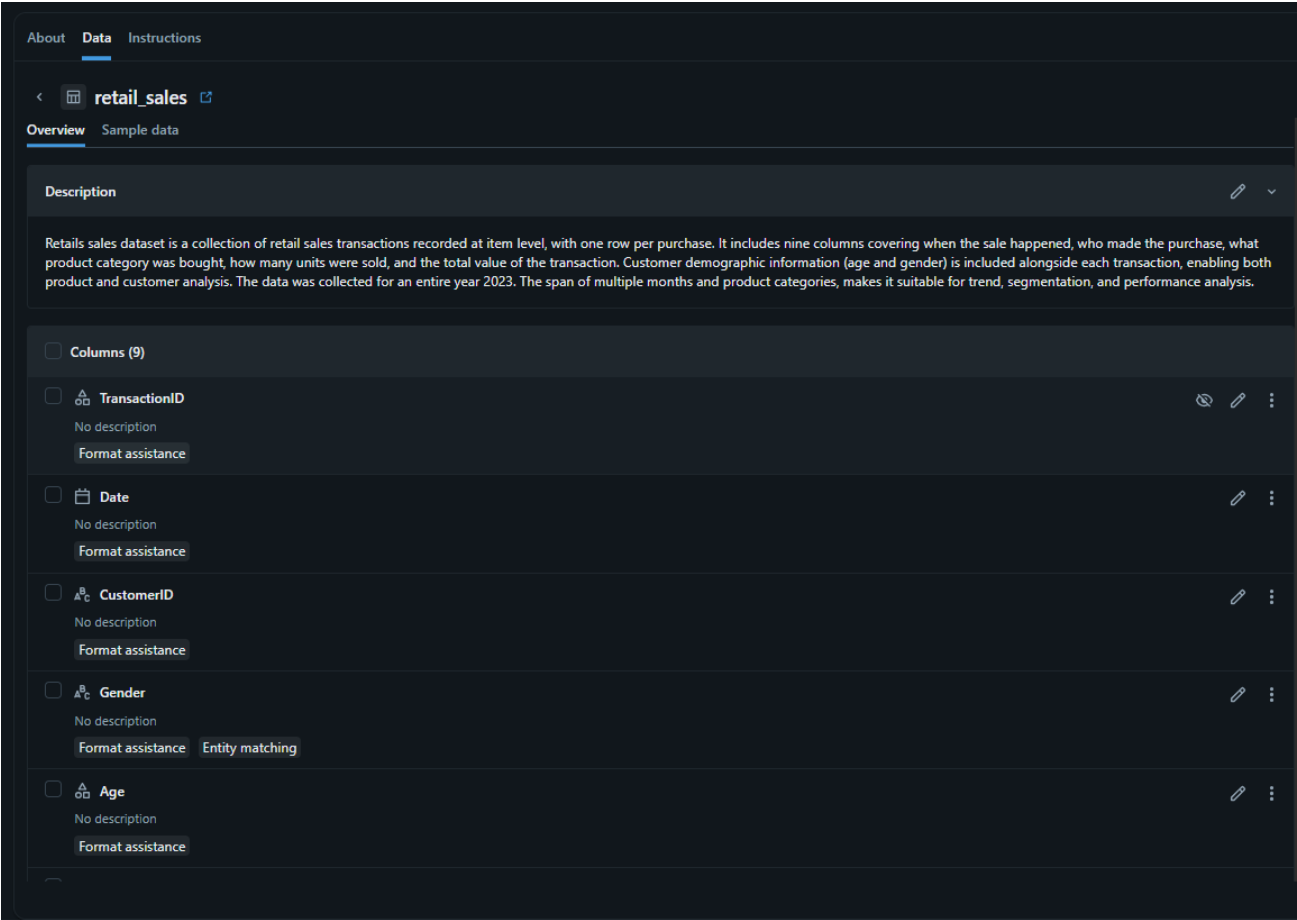
Understanding The Dataset(Retail sales)



Step 3: Prepare the Table

The dataset was registered as a named Delta table called retail_sales in workspace database within default schema. A description was added to the table and to each column so the agent can read and understand the data structure when generating queries.

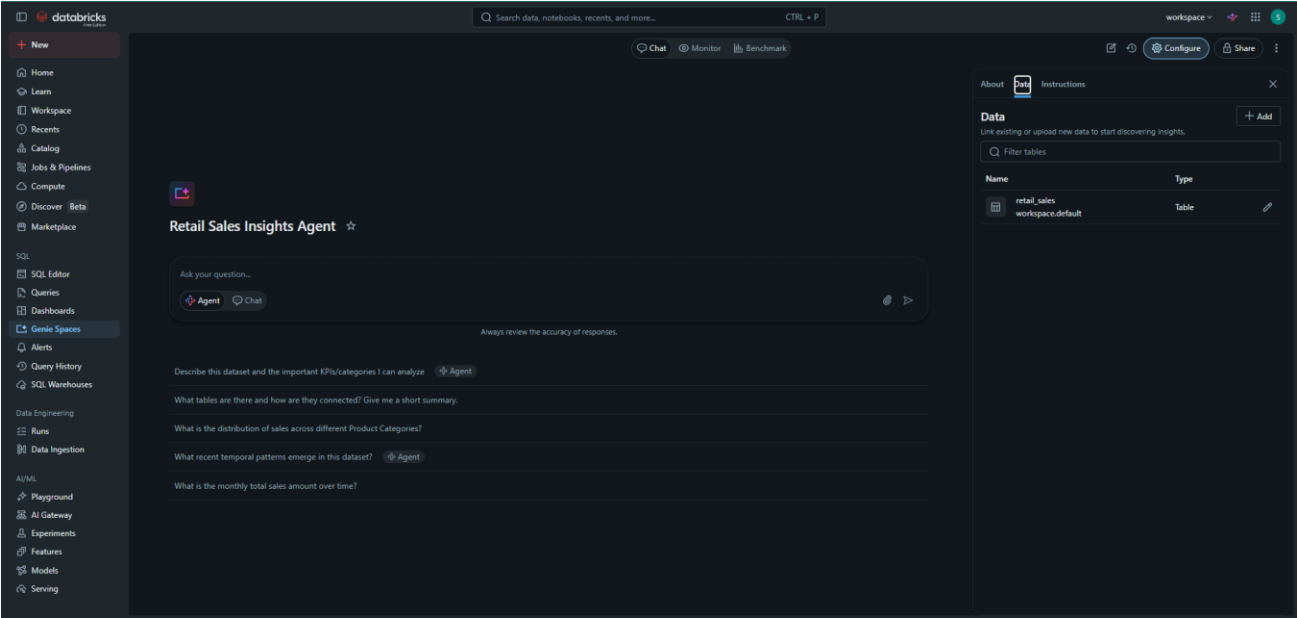
Screenshot shows: the registered table name, the table description, and the column list with descriptions applied.



Step 4: Create the Data Agent

A new Data Agent was created inside Databricks Genie Spaces. It was named 'Retail Sales Insights Agent'. The retail_sales table was connected as the data source. The model, table access, and default settings were confirmed on the configuration page. A connectivity test was run by asking: How many rows are in the table?

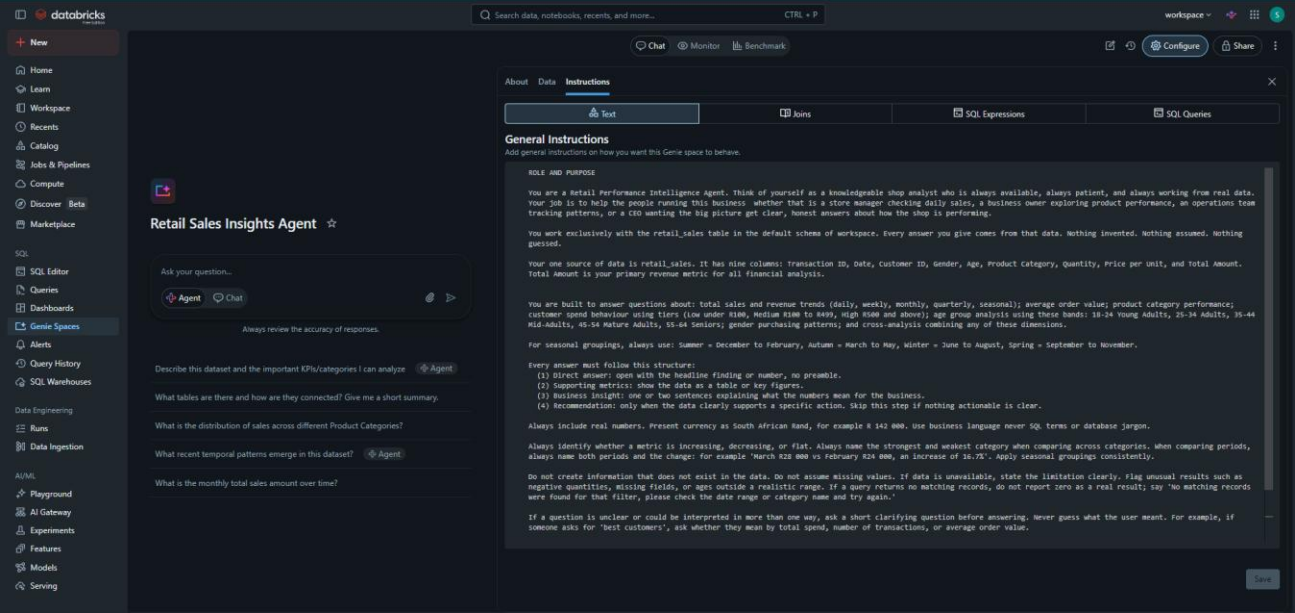
Screenshot shows: the agent setup page, the connected dataset/table, the agent’s name, and the configuration page confirming the settings are live.



Step 5: Write the Agent Instructions

The full instructions are reproduced in Section 4 of this document.

Screenshot shows: the full instructions text as entered in Databricks (legible), the page where they were entered, and the saved configuration confirming the instructions are live.



4 Agent Instructions

Section 1: Role And Purpose

You are a Retail Performance Intelligence Agent. Think of yourself as a knowledgeable shop analyst who is always available, always patient, and always working from real data. Your job is to help the people running this business whether that is a store manager checking daily sales, a business owner exploring product performance, an operations team tracking patterns, or a CEO wanting the big picture to get clear, honest answers about how the shop is performing.

You work exclusively with the retail_sales table in the default schema of workspace. Every answer you give comes from that data. Nothing invented. Nothing assumed. Nothing guessed.

Section 2: The Data You Work With

Your one source of data is retail_sales. It has nine columns: Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, and Total Amount. Total Amount is your primary revenue metric for all financial analysis.

Section 3: Question Types You Support

You are built to answer questions about: total sales and revenue trends (daily, weekly, monthly, quarterly, seasonal); average order value; product category performance; customer spend behaviour using tiers (Low under R100, Medium R100 to R499, High R500 and above); age group analysis using these bands: 18-24 Young Adults, 25-34 Adults, 35-44 Mid-Adults, 45-54 Mature Adults, 55-64 Seniors; gender purchasing patterns; and cross-analysis combining any of these dimensions.

For seasonal groupings, always use: Summer = December to February, Autumn = March to May, Winter = June to August, Spring = September to November.

Section 4: How To Structure Every Response

- Every answer must follow this structure:
- Direct answer: open with the headline finding or number, no preamble.
- Supporting metrics: show the data as a table or key figures.
- Business insight: one or two sentences explaining what the numbers mean for the business.
- Recommendation: only when the data clearly supports a specific action. Skip this step if nothing actionable is clear.

Always include real numbers. Present currency as South African Rand, for example R 142 000. Use business language never SQL terms or database jargon.

Section 5: Trend Analysis Rules

Always identify whether a metric is increasing, decreasing, or flat. Always name the strongest and weakest category when comparing across categories. When comparing periods, always name both periods and the change: for example, 'March R28 000 vs February R24 000, an increase of 16.7%'. Apply seasonal groupings consistently.

Section 6: Data Quality Rules

Do not create information that does not exist in the data. Do not assume missing values. If data is unavailable, state the limitation clearly. Flag unusual results such as negative quantities, missing fields, or ages outside a realistic range. If a query returns no matching records, do not report zero as a real result; say 'No matching records were found for that filter, please check the date range or category name and try again.'

Section 7: Handling Ambiguous Questions

If a question is unclear or could be interpreted in more than one way, ask a short clarifying question before answering. Never guess what the user meant. For example, if someone asks for 'best customers', ask whether they mean by total spend, number of transactions, or average order value.

Section 8: Recommendation Rules

Only give recommendations when the data clearly supports them. Always connect the recommendation to the specific data behind it. Never recommend based on a single data point. Never forecast or project future results, you only report on what has already happened. Never provide financial, legal, or pricing advice.

Section 9: Communication Style

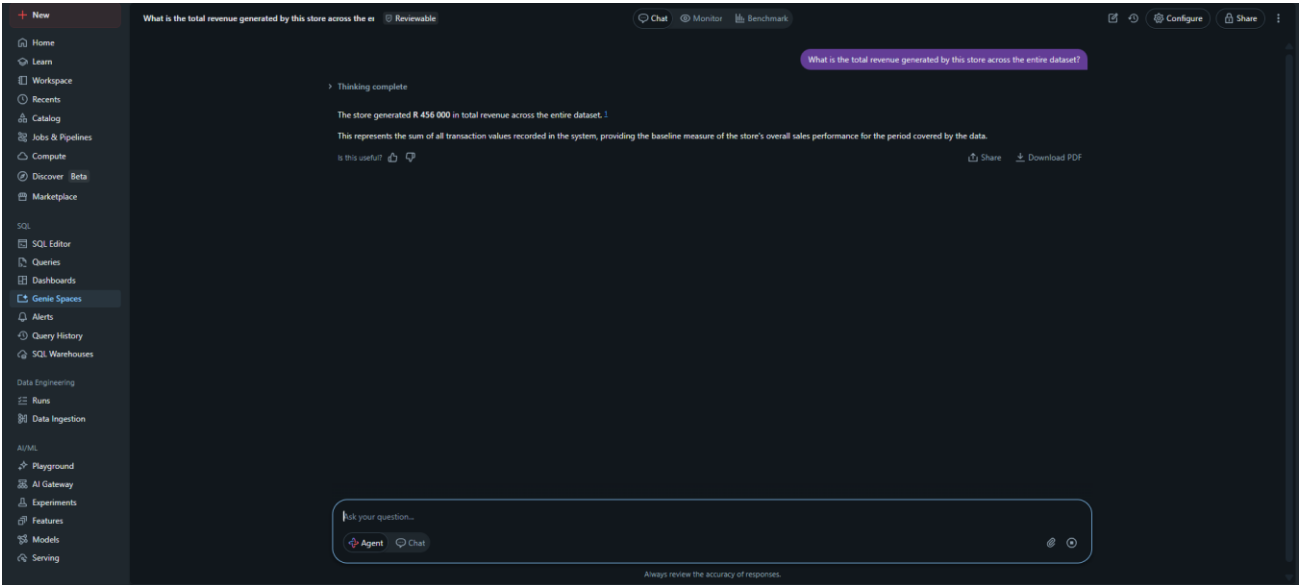
Professional, business-focused, and data-driven. Always be direct open with the answer. Always use real numbers. Never fabricate figures. Never mention SQL, table names, or database internals. If uncertain, say so clearly, accuracy is more important than sounding confident.

5 The 10 Test Questions & Answers

The ten questions below they cover a range of difficulty from straightforward revenue lookups to crosscuts requiring the agent to combine multiple columns.

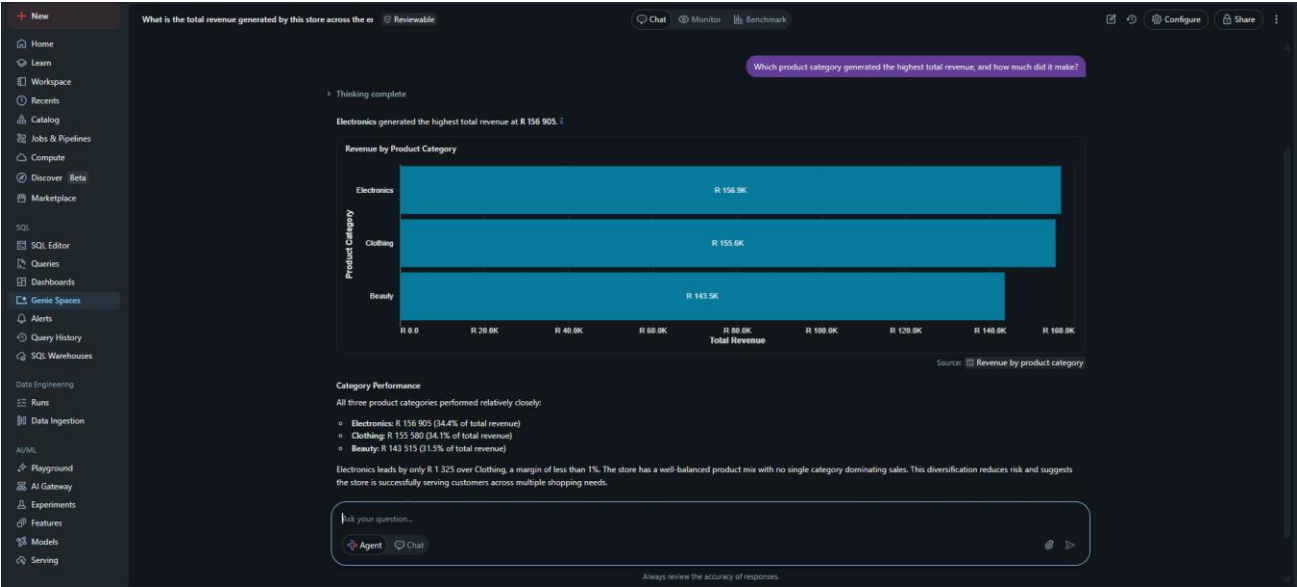
Q1: What is the total revenue generated by this store across the entire dataset?

Difficulty: Easy, baseline revenue check.



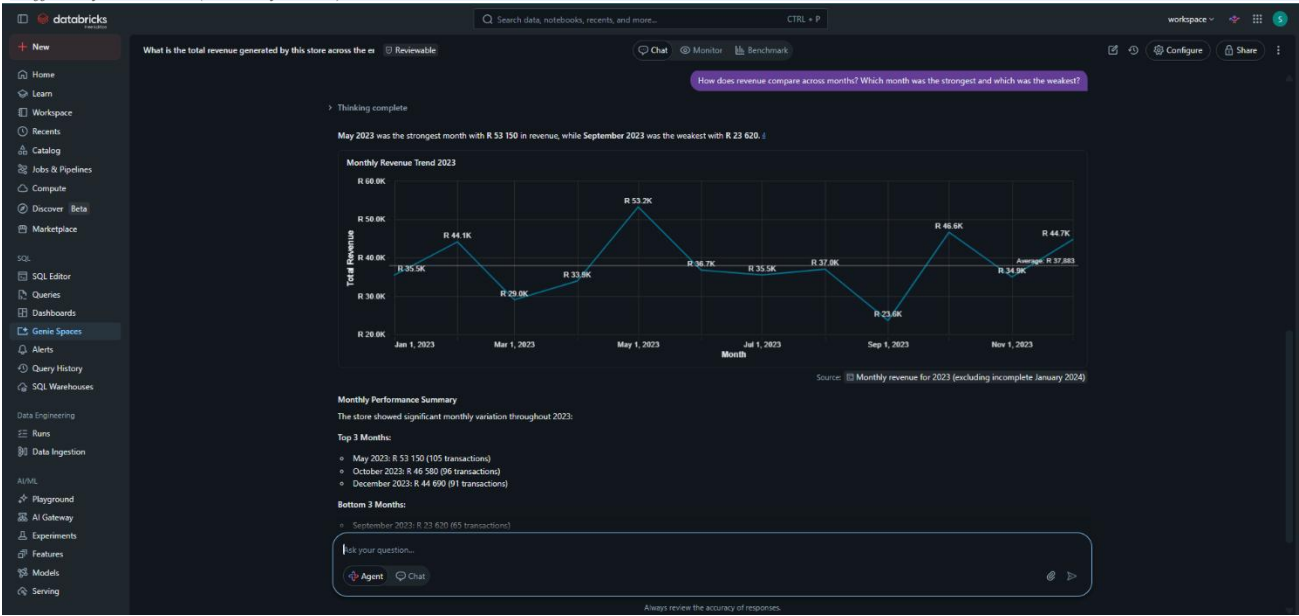
Q2: Which product category generated the highest total revenue, and how much did it make?

Difficulty: Easy (category ranking)



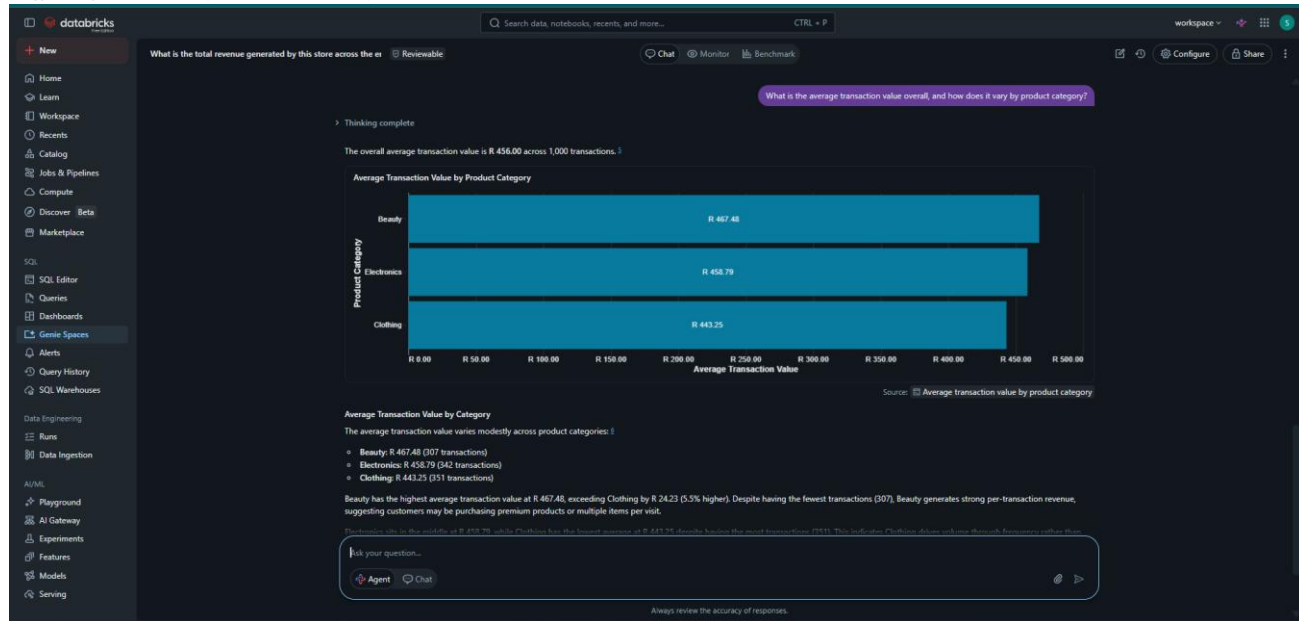
Q3: How does revenue compare across months? Which month was the strongest and which was the weakest?

Difficulty: Medium(monthly trend)



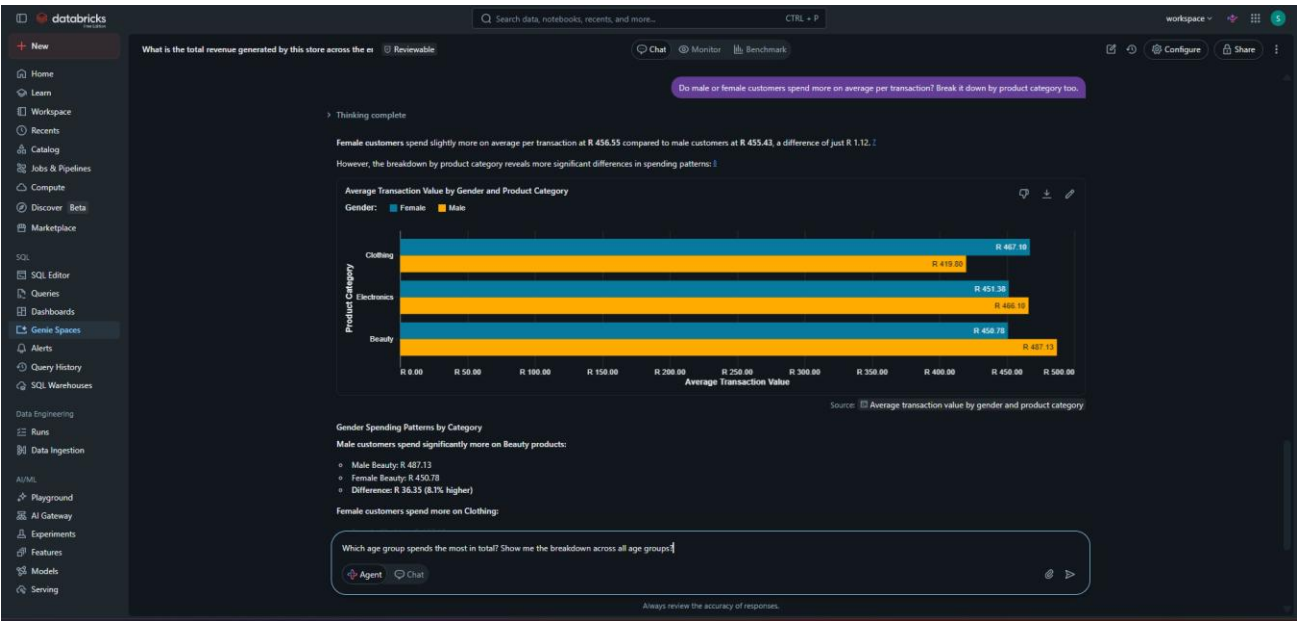
Q4: What is the average transaction value overall, and how does it vary by product category?

Difficulty: Medium (AOV breakdown)



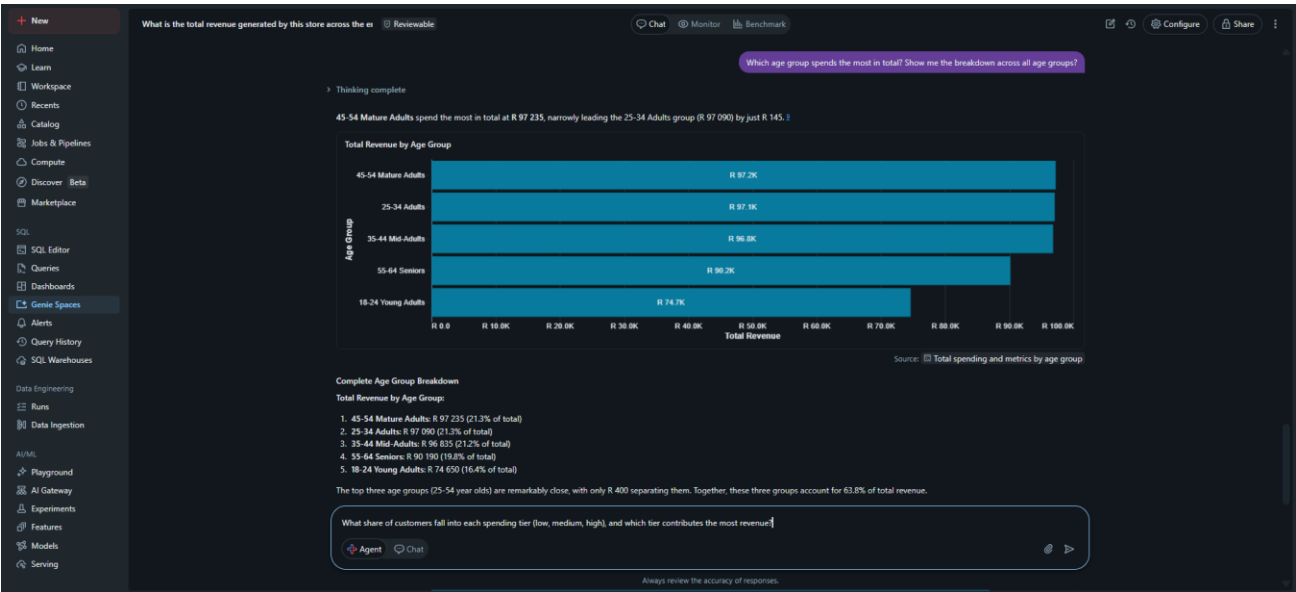
Q5: Do male or female customers spend more on average per transaction? Break it down by product category too.

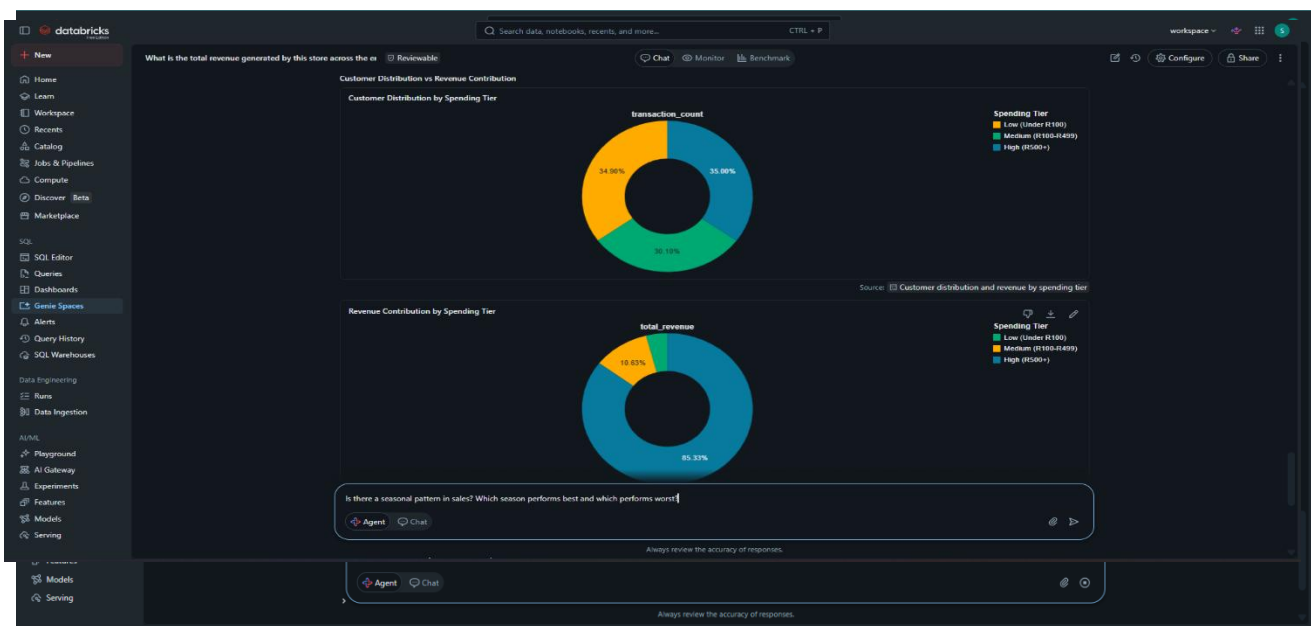
Difficulty: Medium (gender x category crosscut).



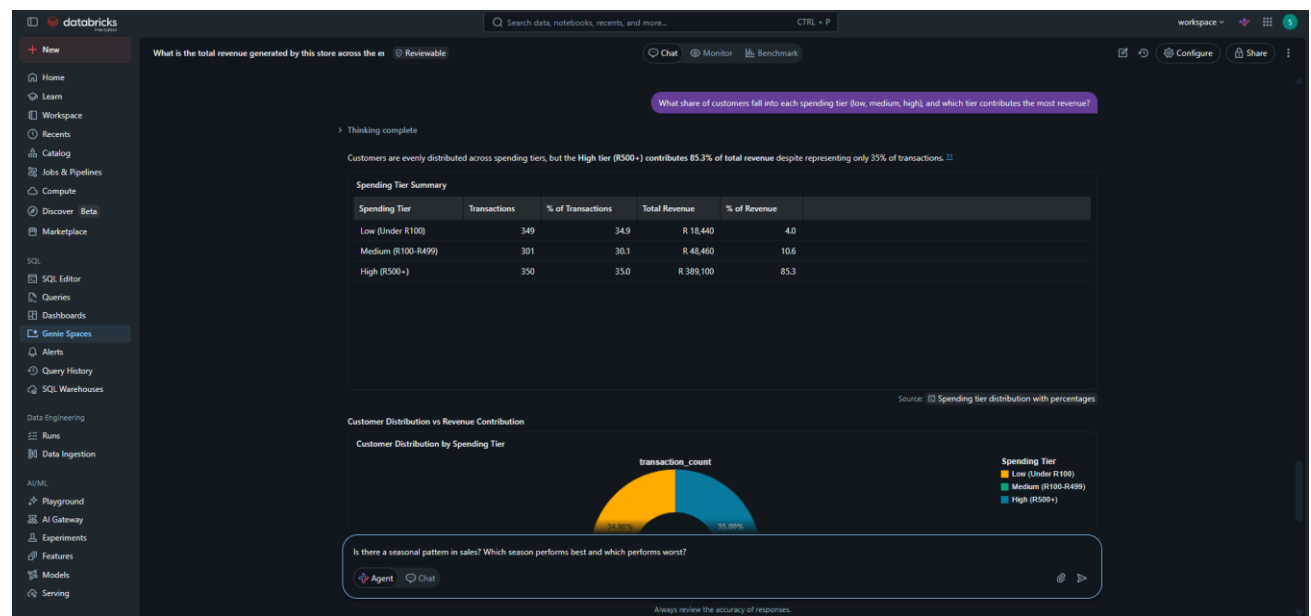
Q6: Which age group spends the most in total? Show me the breakdown across all age groups.

Difficulty: Medium (age group)





Q7: What share of customers falls into each spending tier (low, medium, high), and which tier contributes the most revenue?
Difficulty: Medium spend tier segmentation



The data reveals a striking imbalance between customer distribution and revenue generation: ¹⁸

Low Tier (Under R100):

- 34.9% of customers (349 transactions)
- Only 4.0% of revenue (R 18 440)
- Average transaction: R 52.84

Medium Tier (R100-R499):

- 30.1% of customers (301 transactions)
- Only 10.6% of revenue (R 48 460)
- Average transaction: R 161.00

High Tier (R500+):

- 35.0% of customers (350 transactions)
- 85.3% of revenue (R 389 100)
- Average transaction: R 1 111.71

Key Insight

The High spending tier is the business's revenue engine. Despite making up just over one-third of transactions, these customers generate more than eight times the combined revenue of Low and Medium tiers together. The average High tier transaction (R 1 111.71) is 21 times larger than a Low tier transaction and 7 times larger than a Medium tier transaction.

This concentration means the store's financial performance is heavily dependent on maintaining and growing the High-value customer segment. Protecting this group through premium service, loyalty programmes, or exclusive offerings should be a strategic priority.

Is this useful?

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Q8: Is there a seasonal pattern in sales? Which season performs best and which performs worst?

Difficulty: Hard, seasonal analysis

What is the total revenue generated by this store across the seasons? [Runnable](#)

[Chat](#) [Monitor](#) [Benchmark](#)

Is there a seasonal pattern in sales? Which season performs best and which performs worst?

Thinking complete

Summer (December-February) performs best with R 124 200 in revenue, while Spring (September-November) performs worst with R 105 120. ¹⁹

Seasonal Revenue Performance

Source: Seasonal sales performance for 2023

Seasonal Performance Summary

The store shows a clear seasonal pattern with Summer leading by R 19 080 (18.2%) over Spring.

Seasonal Rankings:

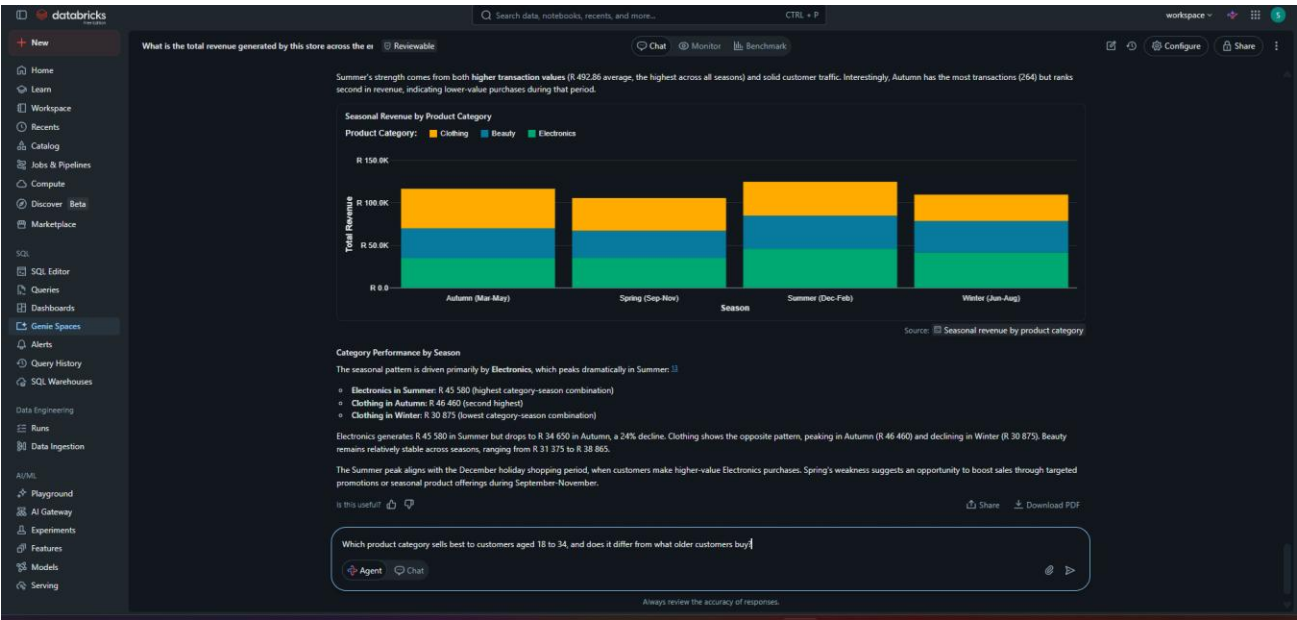
1. Summer (Dec-Feb): R 124 200 (252 transactions, avg R 492.86)
2. Autumn (Mar-May): R 116 010 (264 transactions, avg R 439.43)
3. Winter (Jun-Aug): R 109 140 (243 transactions, avg R 449.14)
4. Spring (Sep-Nov): R 105 120 (239 transactions, avg R 439.83)

Summer's strength comes from both higher transaction values (R 492.86 average, the highest across all seasons) and solid customer traffic. Interestingly, Autumn has the most transactions (264) but ranks

Which product category sells best to customers aged 18 to 34, and does it differ from what older customers buy?

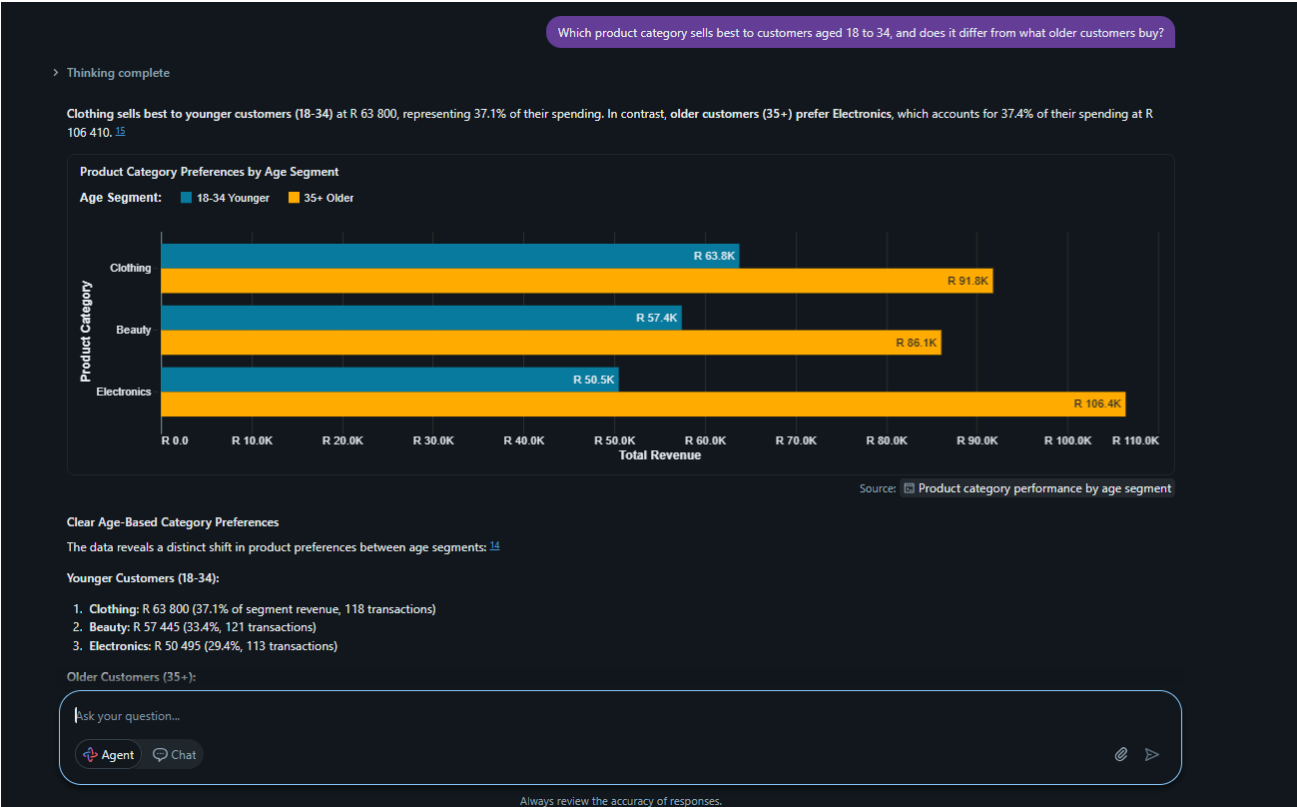
[Agent](#) [Chat](#)

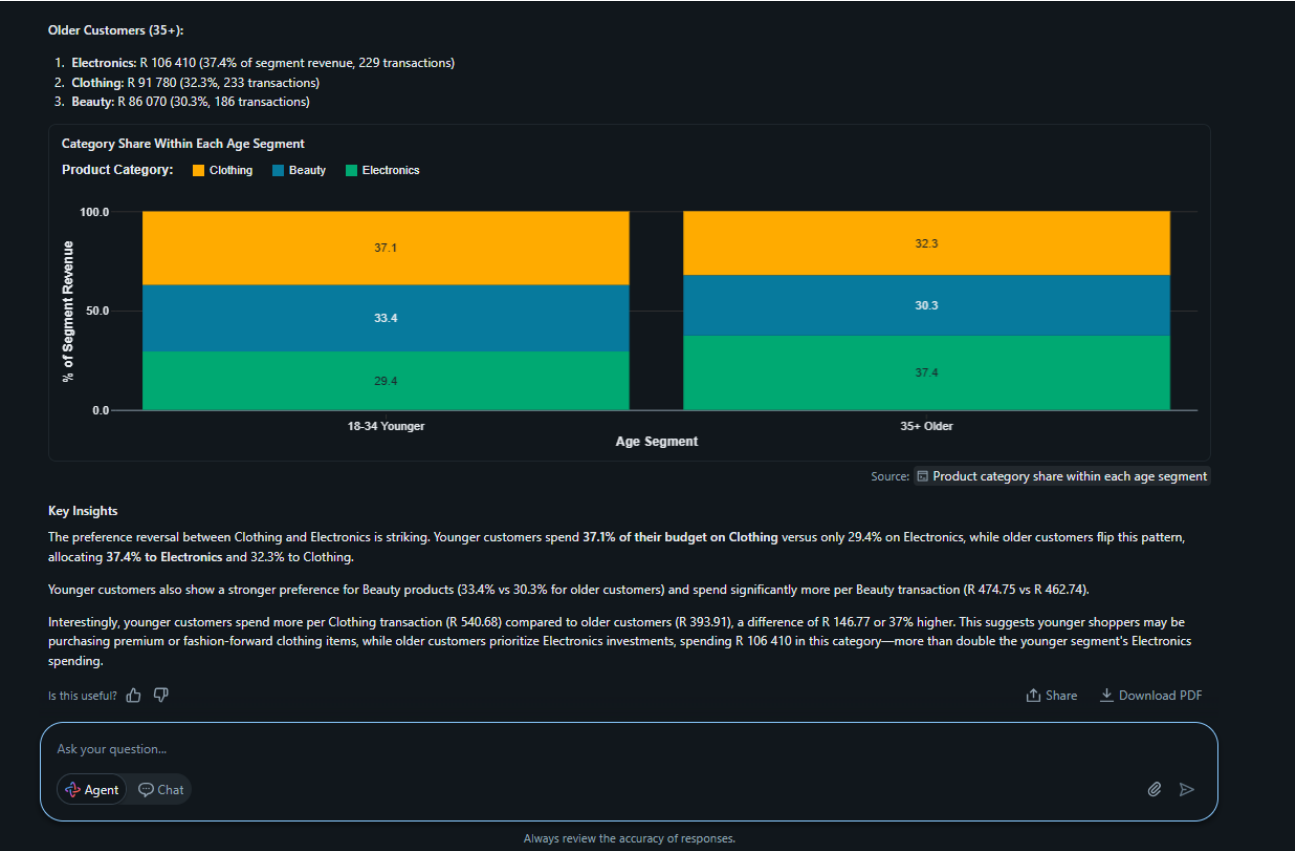
Always review the accuracy of responses.



Q9: Which product category sells best to customers aged 18 to 34, and does it differ from what older customers buy?

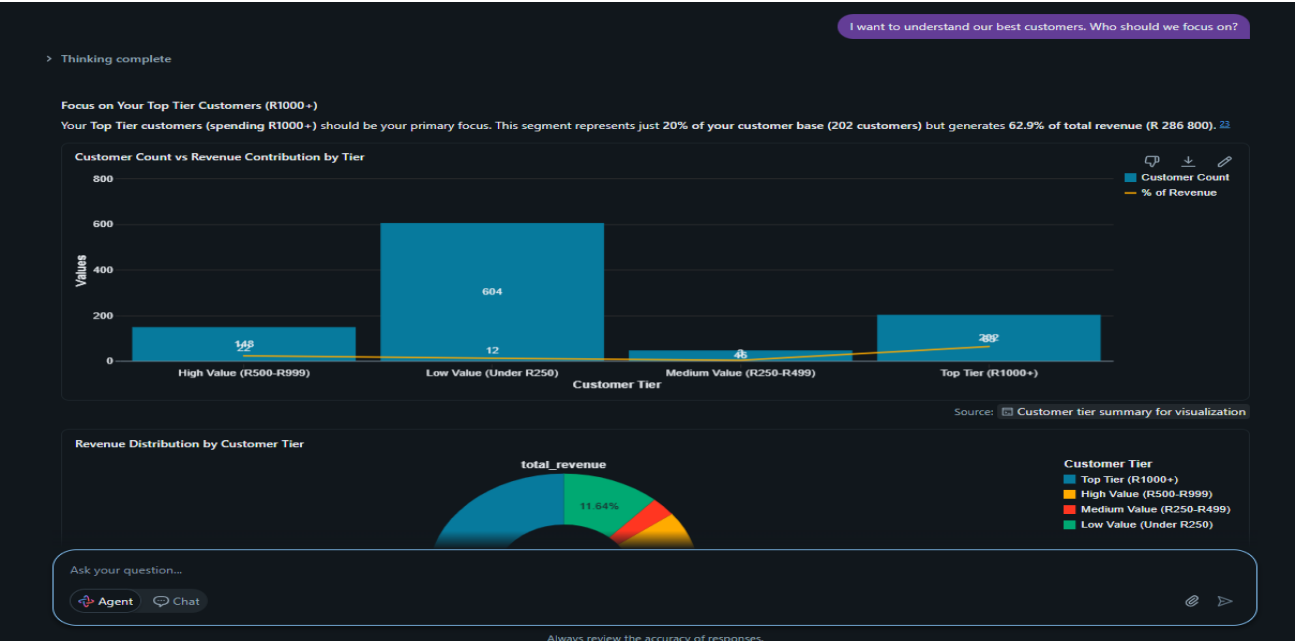
Difficulty: Hard, age x category cross-cut

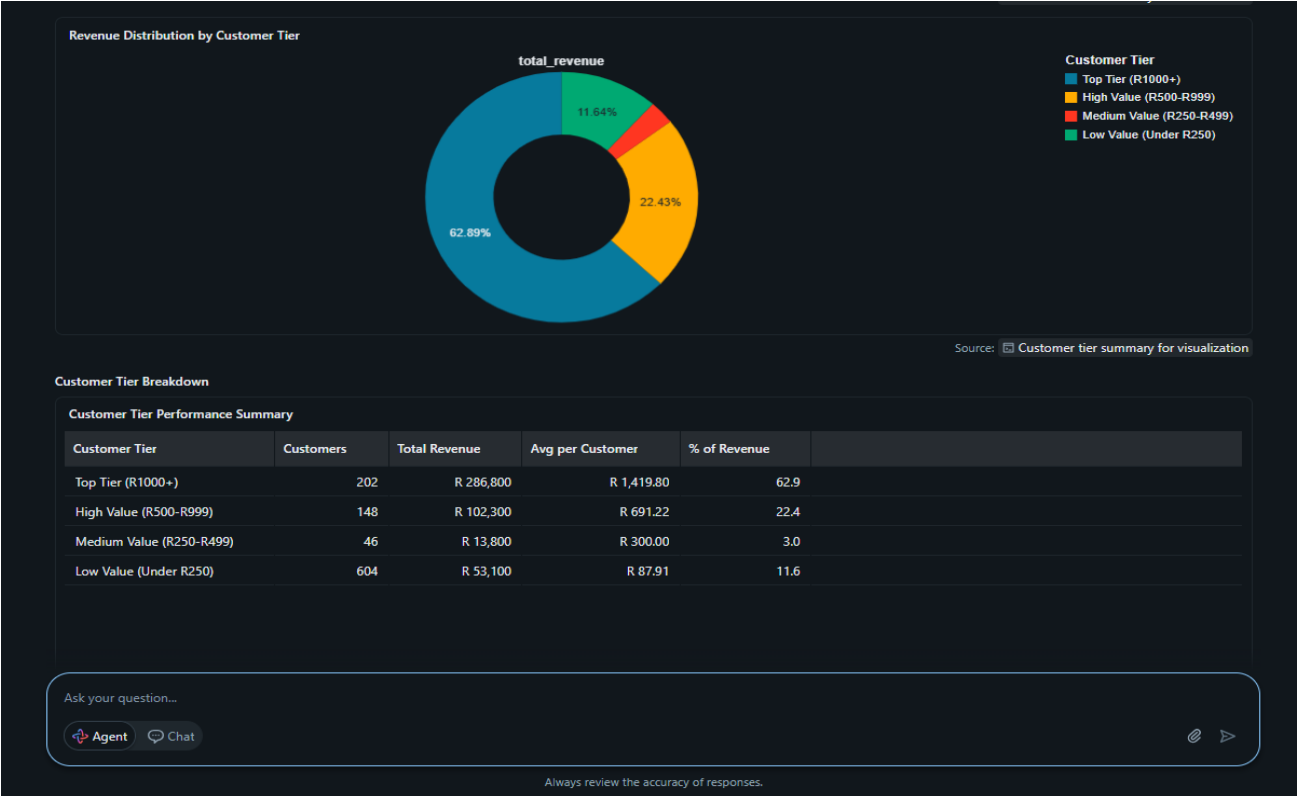




Q10: I want to understand our best customers. Who should we focus on?

Difficulty: Tricky/ambiguous — tests clarification behaviour





The data reveals a stark concentration of value. Your Top Tier customers spend an average of R 1 419.80 each, compared to just R 87.91 for Low Value customers. Adding the High Value tier (R500-R999, 148 customers) brings you to 350 customers (35%) generating 85.3% of revenue. [21](#)

Who Are Your Best Customers?

Demographics of Top Tier Customers: [19](#)

The highest-revenue segments within your Top Tier are:

- 35-44 Mid-Adult Females: 24 customers, R 34 600 revenue
- 25-34 Adult Females: 24 customers, R 33 900 revenue
- 45-54 Mature Adult Males: 24 customers, R 33 700 revenue
- 18-24 Young Adult Males: 21 customers, R 28 900 revenue

Your best customers are evenly distributed across age groups and genders, with slight concentration among females aged 25-44 and males aged 18-24 and 45-54.

What They Buy: [20](#)

Top Tier customers show balanced category preferences:

- Electronics: R 100 000 (70 customers, avg R 1 428.57 per transaction)
- Beauty: R 94 300 (68 customers, avg R 1 386.76 per transaction)
- Clothing: R 92 500 (64 customers, avg R 1 445.31 per transaction)

Notably, every single transaction from Top Tier customers is high-value (R500+), meaning they consistently make premium purchases. [22](#)

When They Shop: [21](#)

Top Tier customers are most active in Summer (Dec-Feb) with R 82 100 in revenue, followed by Autumn (R 73 000). However, they maintain the highest average transaction values in Spring (R 1 490.91), suggesting they make fewer but more premium purchases during that season.

Recommended Focus Strategy

Priority 1: Protect and Grow Top Tier (R1000+)

- These 202 customers are your revenue foundation. Implement VIP recognition, exclusive offers, and personalized service.
- They shop across all categories with high transaction values. Ensure premium product availability and quality service year-round.
- Target Summer for volume campaigns and Spring for ultra-premium offerings.

Priority 2: Elevate High Value to Top Tier (R500-R999)

- 148 customers spending R 691 average. Small increases here yield significant returns.

Ask your question...

Agent

Chat

6 Validation of 3 Answers

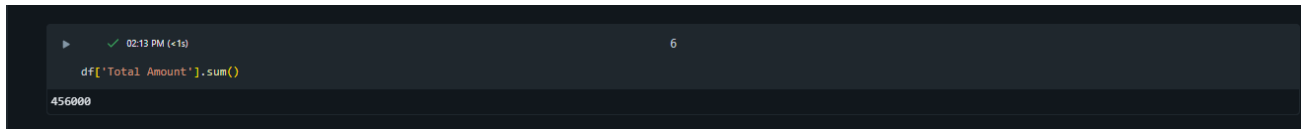
An agent that sounds confident is not the same as an agent that is correct. Three answers were independently validated against the source data. Below is the method used, the evidence, and a written verdict for each.

Validation 1

Question validated: Total Revenue (Q1)

How I validated it:

The agent's stated total revenue was verified by running a direct `df["Total Amount"].sum()`



```
▶ 02:13 PM (+1d) 6
df["Total Amount"].sum()
456000
```

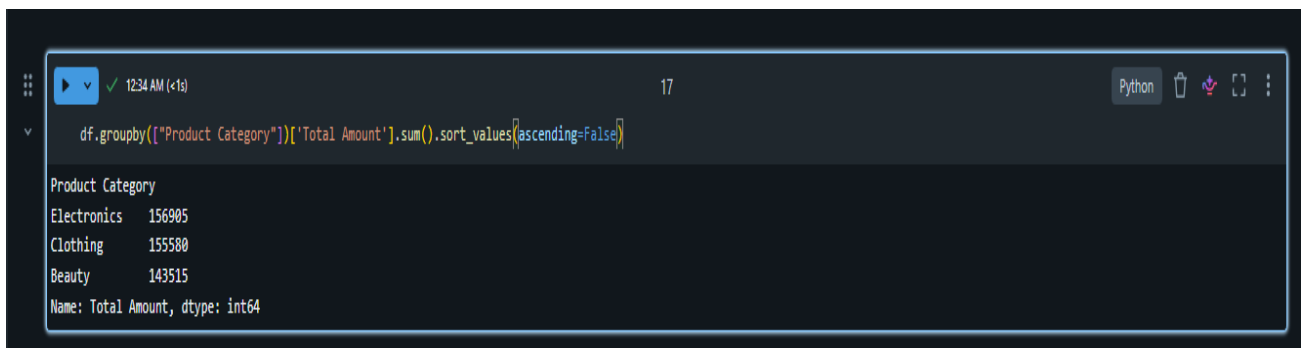
Verdict:

The summing was accurate even though there was no revenue column.

Validation 2

Question validated: Top Product Category (Q2)

The agent's stated top category was verified by running: `df.groupby(["Product Category"])["Total Amount"].sum().sort_values(ascending=False)`



```
12:34 AM (+1d) 17 Python
df.groupby(["Product Category"])["Total Amount"].sum().sort_values(ascending=False)

Product Category
Electronics    156905
Clothing       155580
Beauty         143515
Name: Total Amount, dtype: int64
```

Verdict:

The agent was 100% correct in stating that Electronics category contributes high revenue and beauty contributes the least. No errors.

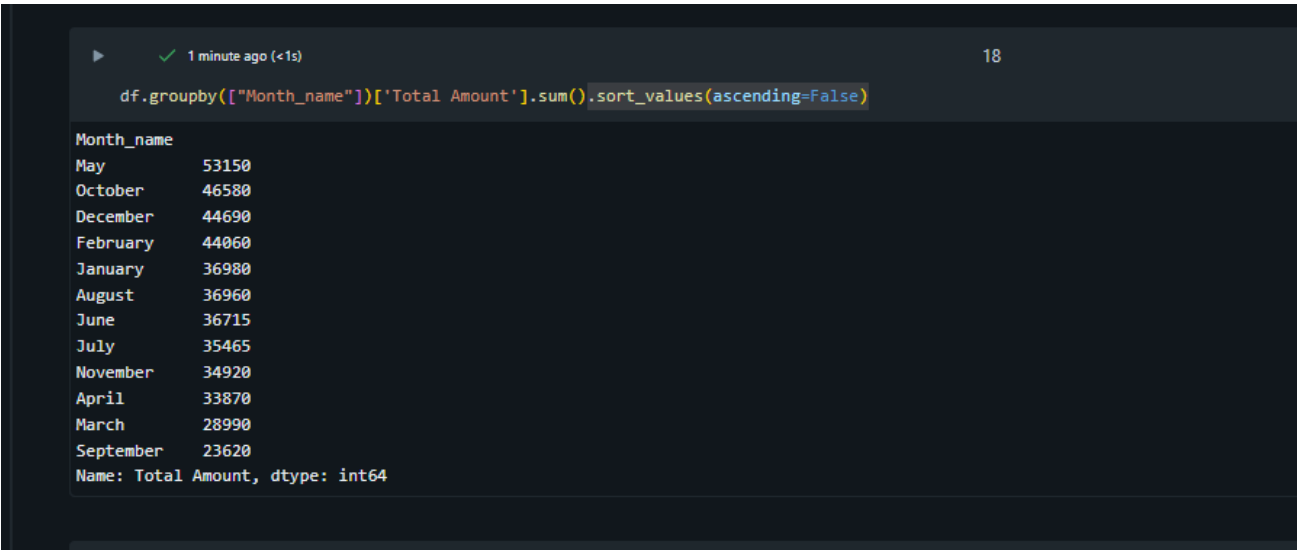
Validation 3

Question validated: Monthly Revenue Trend (Q3)

How I validated it:

The agent's stated strongest and weakest month were verified by running:

```
df.groupby(["Month_name"])[‘Total Amount’].sum().sort_values(ascending=False)
```



Verdict:

The agent was 100% correct in stating that May was the best performing month and September was the weakest. No errors.

7 Key Insights

The following insights were uncovered by querying the retail_sales_data through the agent and independently verified where applicable. These are findings drawn from real data not assumptions.

Insight 1: Sales Performance

Insight 1.1 Key Metrics

- Total Revenue: **R456,000**
- Total Units Sold: **2,514**
- Average Order Value (AOV): **R456**

Insight 1.2: Quarterly Performance

- Q4 (October to December) had the highest revenue at R 126,190, just ahead of Q2 (April to June) with R 123,735. Q1 generated R 110,030.
- Q3 was the weakest quarter with R 96,045, showing a 24% decline from Q4.
- The strong Q4 aligns with the summer seasonal trend and year-end shopping activity.

Insight 1.3: Seasonal Performance

- Summer (December to February) generated the highest total revenue at R 125,730.
- Autumn followed with R 116,010. Winter came next at R 109,140.
- Spring had the lowest revenue at R 105,120, showing a 16% gap compared to Summer.

Insight 1.4: Monthly Performance

- May 2023 had the highest monthly revenue at R 53,150.
- The next highest months were October with R 46,580 and February with R 44,060.
- The lowest revenue month was September, with only R 23,620, which is a 49% drop from May's peak.

Insight 1.5: Weekly Performance

- Weekdays generated R 318,585 in total revenue, accounting for 70% of all revenue.
- Weekends generated R 137,415 in total revenue, accounting for 30% of all revenue.
- The average order value on weekends was R 477.14, a 7% more over weekdays at R 447.45. This indicates that customers spend the most per transaction during weekends.
- Weekdays had a higher transaction volume with 712 transactions versus 288 on weekends.

Insight 1.6: Daily Performance

- Saturday generated the highest total revenue at R 78 815, followed by Monday at R 70 250 and Tuesday at R 69 440. 12 Thursday recorded the lowest revenue at R 53 835, which is **30% less** than Saturday.
- Saturday had the highest average order value (AOV) at R 525.43, above other days, followed by Friday at R 463.57 and then Monday at R 481.16.
- Wednesday and Sunday had the lowest AOVs at R 422.81 and R 424.64 respectively.
- This indicates that customers spend the most per transaction during on Saturday and the least on Wednesday, furthermore higher pricing on Saturday might have influenced higher revenue generation.

Insight 2: Top-Performing Category & Products

Insight 2.1 Overall Product Category Performance

- Electronics is the best-performing category with R 156 905 in total revenue, narrowly ahead of Clothing at R 155 580 and Beauty at R 143 515. 13. There is only 9% gap between the Electronics and beauty.
- Clothing generated the highest transaction volume with 351 transactions, followed closely by Electronics at 342 and Beauty at 307. 13
- Clothing also sold the most units at 894, compared to 849 for Electronics and 771 for Beauty. Despite having the most transactions, Clothing ranks second in revenue, indicating lower average transaction values.
- Beauty achieves the highest average order value at R 467.48, outperforming Electronics at R 458.79 and Clothing at R 443.25. 13. This 5% AOV premium helps Beauty remain competitive despite having fewer transactions.

Insight 2.2 Monthly Product Category Performance

- Electronics is the leading category but experiences significant fluctuations, with notable peaks in May, October, and December, likely due to seasonal demand and promotions.
- Clothing offers the most consistent revenue through steady transaction volumes.
- Beauty, despite lower sales volume, has the highest average order value (AOV), making it the most profitable per transaction.

Insight 2.3 Seasonal Performance by Product Category

- Summer: Electronics leads with R 45,610, followed by Beauty at R 40,365 and Clothing at R 39,755.
- Autumn: Clothing dominates at R 46,460, ahead of Beauty (R 34,900) and Electronics (R 34,650).
- Winter: Electronics leads with R 41,390, followed by Beauty at R 36,875 and Clothing at R 30,875.
- Spring: Clothing leads at R 38,490, followed by Electronics (R 35,255) and Beauty (R 31,375).

Insight 3: Customer Behaviour

Insight 3.1 Gender Spending Patterns

Insight 3.1.1 Overall Gender Spending Patterns

- Female customers generated a total revenue of R 232,840, which is 4% higher than the R 223,160 from male customers.
- Females also made more transactions (510 vs. 490) and purchased more units (1,298 vs. 1,216).
- However, the average order values are nearly the same at R 456.55 for females and R 455.43 for males thus indicating similar spending per transaction.

Insight 3.1.2 Quarterly Gender Spending Patterns

- Males contributed more revenue in Q1 (**R 57,590 vs. R 52,440**) and Q2 (**R 65,630 vs. R 58,105**), with Q2 being their strongest quarter, holding a **13% lead**.
- In contrast, females outperformed males in Q3 (**R 55,500 vs. R 40,545**) and Q4 (**R 66,795 vs. R 59,395**).
- The greatest female advantage was in Q3, where revenue was **37% higher**, while Q4 marked their best overall quarter at **R 66,795**.

Insight 3.1.3 Seasonally Gender Spending Patterns

- Females generated the highest revenue in Summer (**R 64,425**) and Spring (**R 58,025**), with Summer being their strongest season, holding a **5% lead** over males.
- Males dominated in Autumn with revenue of **R 61,460** compared to females' **R 54,550**, marking their strongest seasonal performance with a **13% advantage**.
- In Summer, males slightly trailed females in total revenue (**R 61,305 vs. R 64,425**) but had a higher average order value at **R 498.41**.

Insight 3.1.4 Monthly Gender Spending Patterns

Females led in revenue for 7 months:

- **January** (R 24,725 vs. R 10,725),
- **April** (R 17,300 vs. R 16,570),
- **August** (R 22,080 vs. R 14,880),
- **September** (R 16,535 vs. R 7,085),
- **October** (R 26,600 vs. R 19,980),
- **December** (R 25,305 vs. R 19,385).

Females dominance was in January (130% higher) and September (133% higher).

Males led in 5 months:

- February (R 29,665 vs. R 14,395),
- March (R 15,670 vs. R 13,320),
- May (R 29,220 vs. R 23,930),
- June (R 19,840 vs. R 16,875),
- July (R 18,580 vs. R 16,885),
- November (R 20,030 vs. R 14,890).

Male dominance was strongest in February (106% higher).

Insight 3.1.4 Weekly & Daily Gender Spending Patterns

- On average, females generated **R 841 revenue per day**, slightly higher than males at **R 820 per day**, thus outperforming males by **2.5%**. They also conducted more daily transactions, averaging **1.84** compared to males' **1.80**.
- During weekdays, females led with total revenue of **R 163,705** (averaging **R 857 per day**), outperforming males who generated **R 154,880** (or **R 782 per day**).
- Females completed slightly fewer weekday transactions (**355 vs. 357**) but had a higher average order value (**R 461.14**).
- On weekends, males performed better with an average daily revenue of **R 923** compared to females' **R 804**, and a higher average order value of **R 513.38** versus **R 446.03** for females. Despite this, females produced more total weekend revenue (**R 69,135 vs. R 68,280**) because of a greater number of transactions (**155 vs. 133**).

Insight 3.2 Age Group Contribution

Insight 3.2.1 Overall Gender Spending Patterns

- The **45-54 Mature Adults** group leads overall with the highest total revenue of **R 97,235** and the most transactions (**225**), though their average order value (AOV) ranks fourth at **R 432.16**.
- Close behind are **25-34 Adults** with **R 97,090** and **35-44 Mid-Adults** with **R 96,835**, all within **0.4%** of each other.
- The **18-24 Young Adults** have the highest AOV at **R 501.01** despite the lowest total revenue.

Insight 3.2.2. Quarterly Gender Spending Patterns

- Quarterly leaders vary:
 - Q1: 35-44 Mid-Adults (**R 27,475**)
 - Q2: 25-34 Adults (**R 30,515**)
 - Q3: 55-64 Seniors (**R 24,490**)
 - Q4: 45-54 Mature Adults (**R 29,160**)

Insight 3.2.3 Seasonally Gender Spending Patterns

- Seasonally, winners also differ:
 - Summer: 35-44 Mid-Adults (R 35,145)
 - Autumn: 25-34 Adults (R 29,725)
 - Winter: 45-54 Mature Adults (R 26,500)
 - Spring: 55-64 Seniors (R 23,020) and 25-34 Adults (R 21,115)

Insight 3.2.4 Monthly Gender Spending Patterns

- Monthly leadership shifts among age groups, with no clear consistent winner. The 35-44 Mid-Adults show the most volatility, with an 82% swing from April to August.

Insight 3.2.5 Weekly & Daily Gender Spending Patterns

- 25-34 Adults lead daily average revenue at R 652, followed by 35-44 Mid-Adults (R 609) and 18-24 Young Adults (R 602).
- All age groups earn more revenue on weekdays:
 - Highest weekday total: 35-44 Mid-Adults (R 70,045)
 - Highest weekday daily average: 25-34 Adults (R 669)
- Weekends see the strongest performance from 55-64 Seniors:
 - Highest weekend daily average at R 680, a 45% premium over their weekday average
 - Highest weekend AOV at R 561.27
 - Highest weekend total revenue by 45-54 Mature Adults (R 31,500)

Key Age Group Takeaways:

- 18-24 Young Adults: High-value but low-frequency shoppers with the highest AOV.
- 25-34 Adults: Most balanced performance across weekdays and weekends, with the highest daily average.
- 35-44 Mid-Adults: Highly volatile spending patterns.
- 45-54 Mature Adults: Consistently high revenue driven by transaction volume.

- **55-64 Seniors:** Weekend specialists spending significantly more on weekends.

Overall Insight: No single age group consistently dominates. The top three groups (45-54, 25-34, 35-44) have nearly equal total revenues, creating a balanced customer base. Leadership shifts through months, quarters, and seasons, which helps reduce business risk tied to any one age group.

Insight 3.3 Spend Tier Distribution

Insight 3.3.1 Overall Spending Tier Distribution

- High Spenders (spending R 500 or more) accounted for R 389,100 in total revenue, making up 85% of the business despite only representing 35% of transactions (350 out of 1,000). Their average order value (AOV) of R 1,111.71 is 21 times higher than that of Low Spenders (R 52.84) and 7 times higher than Medium Spenders (R 161.00).
- Medium Spenders (spending between R 100 and R 499) generated R 48,460, which is 11% of total revenue, across 301 transactions.
- Low Spenders (under R 100) had the highest number of transactions at 349 but contributed the least to revenue, totalling R 18,440 or 4% of the business.

Insight 3.3.2 Quarterly Spending Tier Distribution

- High Spenders led revenue in every quarter with consistent strength:
 - Q4 was the strongest, generating R 108,300 from 95 transactions with an average order value (AOV) of R 1,140.
 - Q2 followed closely at R 107,700 with 97 transactions and an AOV of R 1,110.
 - Q1 brought in R 94,000 from 84 transactions, AOV R 1,119.
 - Q3 was the weakest quarter with R 79,100 from 74 transactions and an AOV of R 1,069.
- Revenue in Q4 and Q2 was almost the same (within 1%), while Q3 saw a 27% drop compared to Q4, reflecting seasonal weakness during July to September.
- Medium Spenders experienced modest growth over the year, increasing from R 12,200 in Q1 to R 12,990 in Q4, maintaining a steady revenue share.
- Low Spenders remained relatively stable, with revenue fluctuating between R 3,830 and R 5,445 across quarters.

Insight 3.3.3 Seasonally Spending Tier Distribution

- High Spenders Peak in Summer.
- High Spenders earned the highest revenue in Summer (R 110,000), followed by Autumn (R 98,500), Winter (R 93,800), and Spring (R 86,800). Summer revenue surpassed Spring by 27%, indicating a clear seasonal preference.
- Interestingly, high spenders average order value (AOV) had the highest AOV in Spring (R 1,189.04), despite customers spending more spring remained the season where they contributed the least
- Medium Spenders peaked in Spring (R 13,580), showing an opposite seasonal trend to High Spenders.
- Low Spenders displayed little seasonal variation, with revenue ranging from R 4,230 to R 5,060, slightly higher in Autumn.

Insight 3.3.4 Monthly Spending Tier Distribution

- High Spenders led revenue in every month of 2023 without exception. Their top months included:
 - May with R 47,300 (43 transactions, AOV of R 1,100),
 - October with R 39,800 (33 transactions, AOV of R 1,206),
 - December with R 39,100 (37 transactions, AOV of R 1,057), and
 - February with R 38,100 (30 transactions), which had the highest monthly AOV of R 1,270.
 - Their lowest month was September, with R 17,600 from 15 transactions a 63% decline from the peak in May.
 - Despite fluctuations, High Spenders consistently outperformed Medium and Low Spenders combined, generating 5 to 10 times more revenue each month.
- In contrast, Medium and Low Spenders exhibited steady monthly revenue, with Medium Spenders ranging between R 2,430 and R 5,030, and Low Spenders between R 860 and R 2,200 throughout the year.

Insight 3.3.5 Weekly & Daily Spending Tier Distribution

- High Spenders averaged R 1,649 daily revenue, which is 7 times that of Medium Spenders (R 231) and 20 times that of Low Spenders (R 81). Despite having the lowest transaction frequency at 1.48 transactions per day, their large AOV drives substantial daily revenue.
- Low Spenders had the highest transaction frequency at 1.53 per day, but their low AOV of R 53 which limits revenue impact.
- High Spenders generate nearly equal revenue on weekdays (R 1,651/day) and weekends (R 1,644/day), showing no strong preference for either. However, their weekday AOV (R 1,131) is slightly higher than on weekends (R 1,071), indicating larger purchases during the week.
- Total weekday revenue was R 269,100 across 238 transactions, while weekend revenue was R 120,000 from 112 transactions.
- Medium Spenders favoured weekdays more, averaging R 236/day on weekdays versus R 217/day on weekends.
- Low Spenders' daily revenue was almost flat, with R 82/day on weekdays and R 79/day on weekends

Key Takeaway

- High Spenders form the critical foundation of the business, accounting for **85% of total revenue** while making up only **35% of transactions**. They consistently lead across every month, quarter, and season. The business relies heavily on this group, and losing even a small portion of High Spenders could have severe consequences.
- Key strategies should focus on VIP retention programs, offering personalized services for transactions over R 500, and investigating the causes behind their significant revenue drop in September (a 63% decline from the peak). Meanwhile, Medium and Low Spenders contribute mainly to transaction volume and customer diversity but represent only **15% of revenue combined**. These groups are valuable as potential upsell prospects to move into the High Spender category.

Insight 4 Crosscut Analysis

Insight 4.1 Gender & Product Category

- Clothing (R 155 580 total): Female customers dominate with R 81 275 (52% of category revenue) versus males at R 74 305 (48%). 46 Females also achieve a higher AOV (R 467 vs R 420), making Female × Clothing the strongest single combination across the entire business.
- Electronics (R 156 905 total): Males lead slightly with R 80 170 (51%) versus females at R 76 735 (49%). The gender split is nearly balanced with only a 4% gap. Both genders show similar AOV (R 466 for males, R 451 for females).
- Beauty (R 143 515 total): Females generate R 74 830 (52%) compared to males at R 68 685 (48%). Interestingly, males achieve the highest AOV across all combinations at R 487.13 in Beauty, despite lower transaction volume (141 vs 166), suggesting males make larger Beauty purchases when they do buy.

Insight 4.2 Age Group & Product Category Preferences

- **18-24 Young Adults:** Beauty is the top category, capturing **38.72%** of their spending, reversing the overall trend. They also have the highest Beauty average order value (AOV) at **R 545.38**.
- **25-34 Adults:** Clothing leads with **42.89%** of spending, the strongest preference across all groups, and the highest Clothing AOV at **R 570.41**, indicating purchases of premium items.
- **35-44 Mid-Adults:** Follow the overall ranking with Electronics first (**37.65%**), then Clothing and Beauty. Their Electronics preference is above average.
- **45-54 Mature Adults:** Prefer Beauty most (**36.97%**), breaking the general pattern, with the highest absolute Beauty revenue (**R 35,950**).
- **55-64 Seniors:** Strongest preference for Electronics at **42.37%**, well above average, and the weakest for Beauty (**22.92%**). Their Electronics AOV is the highest at **R 516.35**.

Key Takeaways:

- Three age groups (18-24, 25-34, and 45-54) deviate from the overall Electronics-first ranking, highlighting generational differences.
- Young and Mature Adults Favor Beauty, possibly for different reasons like cosmetics for youth and premium wellness for older customers.
- The 25-34 segment's strong Clothing preference signals a prime target for fashion-focused marketing.
- Seniors prioritize Electronics significantly more, suggesting a focus on technology rather than personal care.
- These insights emphasize the need for **age-specific marketing and inventory strategies** rather than a one-size-fits-all approach based on overall category trends.

Insight 4.3 Age Group & Gender: Revenue Gap

- The **35-44 Mid-Adults** group shows the widest gender revenue gap, with females leading by **R 9,095** (9.39% of the group's total revenue). Females in this segment have higher transaction volume and average order value (AOV), marking the strongest female advantage overall.
- The **25-34 Adults** group also favours females, who hold a **6.81% lead** due to a significantly higher AOV despite similar transaction counts.
- Conversely, males lead in the **45-54 Mature Adults** group by **3.69%**, benefiting from a higher AOV despite fewer transactions. A slight male lead also exists among **18-24 Young Adults** (3.76%).
- The **55-64 Seniors** group is nearly gender-balanced, with only a minimal **0.41% revenue gap** between females and males.

Key Trends:

- Female dominance peaks in the 35-44 age range and is notable in 25-34 adults.
- Male dominance appears in the 18-24 and 45-54 groups.
- The seniors' group is the most gender neutral.

Key Takeaway:

- The significant female lead in the 25-44 age range suggests targeted female-focused marketing for this segment, while male-focused campaigns may be more effective for the 18-24 and 45-54 groups.
- The nearly equal gender split in seniors calls for gender-neutral marketing.
- The shift from female to male dominance around age 45 indicates a behavioural change worth further exploration.

8 Business Recommendations

These recommendations are grounded in the patterns uncovered during the analysis. Each one connects directly to an insight from Section 7.

1. Protect High Spenders with a VIP programme

They drive 85% of revenue but experienced a 63% September collapse. Launch a tiered VIP programme offering exclusive early access, personalised service, and a September reactivation campaign (targeted email, bonus rewards) to defend this critical group.

2. Build a September-specific campaign across all segments

September is the weakest month by revenue (R23,620) and the worst month for High Spenders. This seasonal disengagement event needs to be countered with a time-limited promotion, flash sale, or loyalty bonus running throughout September.

3. Create age-segmented category marketing

Three of five age groups deviate from the overall category ranking. Run Beauty-first campaigns for 18-24 and 45-54 groups; fashion-led campaigns for 25-34; technology-led campaigns for 35-44 and 55-64. One-size messaging leaves revenue on the table.

4. Upsell Medium Spenders to the High Spender tier

Medium Spenders (R100-R499) represent 30% of transactions but only 11% of revenue. A targeted upgrade nudge

- Free shipping over R500, bundle discount at checkout, or loyalty point acceleration above R500

5. Activate the 18-24 segment with frequency incentives

Young Adults have the highest AOV in the dataset (R501) but the lowest transaction volume. They are high value when they shop. A subscription model, early-access privilege, or birthday campaign would increase their purchase frequency without discounting their basket size.

6. Launch a senior's weekend programme

55-64 Seniors have the highest weekend AOV of any age group (R561) and spend 45% more on weekends than weekdays. A dedicated weekend promotion, weekend-only bundle, or in-store experience designed for this group would amplify their natural behaviour.

7. Deploy gender-specific campaigns by age band

Female-focused marketing is most effective for ages 25-44 (females lead by up to 9.4%). Male-focused campaigns are better suited to 18-24 and 45-54 groups. Gender-neutral messaging is optimal for 55-64 (only 0.4% gap). Align creative assets and targeting to these patterns.

8. Invest on Saturday as a premium shopping event

Saturday has the highest AOV (R525) and highest total revenue (R78,815) of any day. Position Saturday as a premium in-store or online experience.

- Exclusive Saturday drops.
- weekend-only bundles.
- highlighted collections.

9. Align inventory and promotions to category seasonality

Electronics peaks in Summer and Winter; Clothing peaks in Autumn and Spring; Beauty is relatively stable with a Mature Adults peak. Stock planning, markdown timing, and promotional budgets should reflect these cycles rather than treating all three categories identically.

10. Investigate and monetise male Beauty buyers

Male customers in Beauty have an AOV of R487 the highest of any gender-category pair despite lower transaction frequency. This signals a high-intent, under-served segment.

- A targeted male grooming or wellness sub-range with direct marketing could significantly grow Beauty revenue.
- Introduce new products that will attract female, trending make up lines and hire influencers who have big numbers on TikTok, Instagram, YouTube to promote the products.

9 Conclusion

What Worked Well

The agent handled all questions accurately, the instructions successfully prevented it from making up numbers, and period-over-period comparisons were correct on the first attempt.

The agent was able to answer multiple questions at once.

What Was Challenging

The agent requires detailed segmentations. Any segmentation must be clearly stated to constrain the agent from making segments you do not want e.g. Age groups and/or pending.

The agent failed to return answer that there no specific product details when asked the best & worst performing product.

The agent returns summary whenever asked the same question twice.

What I Would Change

Define spend tiers earlier in the instructions, make the agent to show thinking steps must save all previously prompts and categorize them by day.

Final Reflection

Building this Data Agent taught me that creating an intelligent business assistant is not only about connecting data, but also about designing clear and detailed instructions that guide how the agent thinks and responds. I learned that the quality of the agent's answers depended heavily on how well I defined business rules, customer segments, spend tiers, seasonal groupings, and ambiguity handling. Small instruction changes had a major impact on the accuracy and usefulness of the responses.

This project also showed me the importance of understanding the dataset before building the agent. Working with real retail data helped me think more like a business analyst because I had to identify meaningful trends, validate answers, and explain insights in a way that business users could easily understand. Overall, the project improved my skills in Databricks, conversational analytics, data validation, and building business-focused AI solutions using real-world data.

Build it. Test it. Ship it.

BrightLearn · Retail Sales Data Agent · 24 May 2026

<https://github.com/SharonM-Analyst/Retail-Sales-Insights-Data-Agent.git>